

PRINTING PATTERN IN PYTHON

1.print rectangle

```
In [2]: print('# # # #')
        print('# # # #')
        print('# # # #')
        print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [6]: # 1st way
        for i in range(4):
            for j in range(4):
                print('#',end=' ')
            print()           # print function for next line
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [9]: # 2nd way
        for i in range(1,5):
            print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [10]: #3rd way
         for i in range(1,5):
```

```
if i<=5:
    print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [11]: for j in range(4):
          print('#', end=" ")

          for j in range(4):
              print('#', end=" ")
```

```
# # # # # # # #
```

```
In [12]: for j in range(4):
          print('#', end=" ")

          print()

          for j in range(4):
              print('#', end=" ")
```

```
# # # #
# # # #
```

2.Right-angled triangle

```
In [18]: for i in range(6):
          for j in range(i+1):
              print('#',end=' ')
          print()
```

```
#
# #
# # #
# # # #
# # # # #
# # # # # #
```

```
In [20]: for i in range(1,6):  
        print("# "*i)
```

```
#  
# #  
# # #  
# # # #  
# # # # #
```

3. Inverted Right angled triangle

```
In [21]: for i in range(4):  
        for j in range(4-i):  
            print('#', end=" ")  
        print()
```

```
# # # #  
# # #  
# #  
#
```

```
In [23]: for i in range(1,5):  
        print(" # "*(5-i))
```

```
# # # #  
# # #  
# #  
#
```

4. Pyramid

```
In [1]: for i in range(1,6):  
        print('*'(5-i) + ' # '*(2*i-1) )
```

```
#
# # #
# # # # #
# # # # # # #
# # # # # # # #
```

```
In [58]: for i in range(1,6):
        for j in range(1,10):
            if ((j>=6-i) and (j<=4+i)):
                print(' * ',end=' ')
            else:
                print('   ',end=' ')

        print()
```

```
      *
    * * *
  * * * * *
* * * * * * *
```

5. Inverted Pyramid

```
In [16]: for i in range(5,0,-1):
        print('*'(5-i)+' # '*'(2*i-1))
```

```
# # # # # # # #
# # # # # #
# # # #
# #
#
```

```
In [16]: for i in range(1,6):
        for j in range(1,10):
            if ((j>=i) and (j<=10-i)):
                print(' * ',end=' ')
            else:
                print('   ',end=' ')

        print()
```

```
print()
```

```
* * * * *
  * * * *
    * * *
      * *
        *
```

6.Diamond Pattern

```
In [20]: for i in range(1,6):
          for j in range(1,10):
              if ((j>=6-i) and (j<=4+i)):
                  print(' * ',end=' ')
              else:
                  print('   ',end=' ')

          print()

          for i in range(2,6):
              for j in range(1,10):
                  if ((j>=i) and (j<=10-i)):
                      print(' * ',end=' ')
                  else:
                      print('   ',end=' ')

          print()
```

```
          *
        * * *
      * * * * *
    * * * * * *
  * * * * * * *
* * * * * * * *
  * * * * *
    * * *
      * *
        *
```

7.Hallow Square Pattern

```
In [32]: for i in range(5):
          for j in range(5):
              if(i==4 or i==0 or j==4 or j==0):
                  print('*',end = ' ')
              else:
                  print(' ',end = ' ')

          print()
```

```
* * * * *
*         *
*         *
*         *
*         *
* * * * *
```

8.Full Square Pattern

```
In [33]: for i in range(5):
          for j in range(5):
              print('*',end = ' ')

          print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

9.Right angled triangle(number pattern)

```
In [26]: for i in range(5):
          for j in range(i):
```

```

        print(i,*(' '),end = ' ')
    print()

```

```

1
2 2
3 3 3
4 4 4 4

```

10. Inverted Right angled triangle(number pattern)

```

In [29]: for i in range(6):
        for j in range(5-i):
            print(i,*(' '),end = ' ')
        print()

```

```

0 0 0 0 0
1 1 1 1
2 2 2
3 3
4

```

11. Hallow Right angled triangle

```

In [37]: for i in range(1,6):
        for j in range(1,i+1):
            if (j==1 or i==5 or j==i):
                print(' * ',end='')
            else:
                print(' ',end='')

        print()

```

```

*
* *
*   *
*     *
* * * * *

```

12.Hallow Pyramid pattern

```

In [51]: for i in range(1,6):
          for j in range(1,10):
              if(i==5 or j==6-i or j==4+i):
                  print(' * ',end='')
              else:
                  print('   ',end='')

          print()

```

```

      *
    * *
  *   *
*     *
* * * * *

```

13.Hallow Diamond Pattern

```

In [55]: for i in range(1,5):
          for j in range(1,8):
              if(j==5-i or j==3+i):
                  print(' * ',end='')
              else:
                  print('   ',end='')

          print()

          for i in range(2,5):
              for j in range(1,8):

```

```

    if(j==i or j==8-i):
        print(' * ',end='')
    else:
        print('   ',end='')

print()

```

```

      *
     * *
    *   *
   *     *
  *       *
 *         *
*           *
 *         *
  *       *
   *     *
    *   *
     * *
      *

```

14.Hallow Diamond Pattern(number pattern)

```

In [56]: for i in range(1,5):
        for j in range(1,8):
            if(j==5-i or j==3+i):
                print(f' {i} ',end='')
            else:
                print('   ',end='')

        print()

for i in range(2,5):
    for j in range(1,8):
        if(j==i or j==8-i):
            print(f' {i} ',end='')
        else:
            print('   ',end='')

    print()

```

```

      1
    2 2
  3   3
4     4
  2   2
    3 3
      4

```

15.Butterfly pattern

```

In [79]: for i in range(1,5):
          for j in range(1,8):
              if((j>=1 and j<=i) or (j>=8-i and j<=7)):
                  print(' * ',end='')
              else:
                  print('   ',end='')

          print()

          for i in range(1,5):
              for j in range(1,8):
                  if((j>=1 and j<=5-i) or (j>=3+i and j<=7)):
                      print(' * ',end='')
                  else:
                      print('   ',end='')

              print()

```

```

*           *
*  *       *  *
*  *  *   *  *  *
*  *  *  *  *  *  *
*  *  *  *  *  *  *
*  *  *   *  *  *
*  *       *  *
*           *

```

16.Hallow Number Pyramid

```
In [57]: for i in range(1,6):
          for j in range(1,10):
              if(i==5 or j==6-i or j==4+i):
                  print(f' {i} ',end='')
              else:
                  print(' ',end='')

          print()
```

```
      1
    2   2
  3       3
4   4       4
5 5 5 5 5 5 5 5
```

17.Full star Pyramid

```
In [59]: for i in range(1,6):
          for j in range(1,10):
              if ((j>=6-i) and (j<=4+i)):
                  print(' * ',end='')
              else:
                  print(' ',end='')

          print()
```

```
      *
    * * *
  * * * * *
* * * * * * *
```

18. Inverted full star pyramid

```
In [60]: for i in range(1,6):
          for j in range(1,10):
              if ((j>=i) and (j<=10-i)):
                  print(' * ',end=' ')
              else:
                  print('   ',end=' ')

          print()
```

```
* * * * *
  * * * *
    * * *
      * *
        *
          *
```

19. Left Aligned Pyramid

```
In [62]: for i in range(6):
          for j in range(i+1):
              print('*',end=' ')
          print()
```

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

20. Right Aligned Pyramid

```
In [81]: for i in range(1,6):
          for j in range(1,6):
```

```

        if(j>=6-i and j<=5):
            print(' * ',end=' ')
        else:
            print('   ',end=' ')

    print()

```

```

          *
        * *
      * * *
    * * * *
  * * * * *

```

21.Floyd's triangle

```

In [75]: k=1
for i in range(1,6):
    for j in range(i):
        print(f' {k} ',end=' ')
        k=k+1

    print()

```

```

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

```

In []: